FastSpeech 2

English and Telugu Assignment - Technical Progress Report

Status as of 7 November 2026

Executive summary

This report documents completed, reproducible work and the current blockers. It is a progress report, not a claim that the assignment is complete. The English Svarah corpus was accessed, exported locally, and audited into duration-targeted train and validation splits. Telugu data have not been obtained or validated. Neither language has completed forced alignment, acoustic feature extraction, model training, or inference; therefore no model quality, loss curves, generated speech, or variance plots are reported.

Verified English data

Measure	Observed
Source	ai4bharat/Svarah, source split 'test'
Pinned revision	ebbf7777fe771490696a3f7b007097606fa8c924
Export	6,656 WAV utterances; 9.6144 measured audio hours
Training split	3,429 utterances; 5.0007 hours
Validation split	686 utterances; 1.0001 hours
Overlap / speakers	0 shared IDs; Female and Male labels in both splits
Transcript handling	Split audit left transcript strings unchanged. Correctness has not been manually verified.

The audit is recorded in `data/Svarah/splits/audit.json`; the export metadata is in `data/Svarah/export_summary.json`. Raw audio and manifests are local and excluded from version control. The Svarah dataset card identifies CC BY 4.0; see References.

Implementation and reproducibility

The repository vendors the MIT-licensed ming024/FastSpeech2 reference implementation at commit d4e79eb52e8b01d24703b2dfc0385544092958f3. Language-specific configuration and wrapper scripts, a dataset exporter, and a deterministic duration-based manifest audit are present. The English split uses seed 42. The audit checks audio durations, duplicate IDs, missing files/transcripts, target duration, and train/validation ID overlap.

Environment checks found PyTorch 2.14.1+cpu with CUDA unavailable. The English preprocessing attempt did not reach audio preparation: initially required runtime dependencies were missing, and a subsequent attempt was stopped after it remained without visible progress. Montreal Forced Aligner (MFA) is not installed. No TextGrid alignments or acoustic features were produced. Some Python dependencies were installed into the project-local .venv; the system-wide Python environment was not intentionally modified.

Assignment status

Requirement	Status	Evidence / remaining work
English corpus and 5 h / 1 h split	Complete	Verified Svarah export and disjoint split audit.
Telugu corpus and split	Not started	A public mirror has conflicting language metadata; original IIT Madras access/license process remains unresolved. Do not assume the mirror is Telugu.
Text normalization, lexicon, forced alignment	Not complete	Requires working language-specific frontend and MFA; Telugu pronunciation resources need validation.
FastSpeech 2 training and checkpoints	Not started	Train separate language models; record configuration, hardware, and actual training/validation losses.
Inference, five test cases per language, plots	Not started	Requires trained checkpoints; preserve assignment test strings exactly and render duration, pitch, and energy plots.
Final technical report and public repository	Not complete	This progress report is not the final report. The workspace has no configured Git remote or public repository URL.

Next steps

1. Obtain the original Telugu corpus through an authorized route, review its license, and verify transcript language, audio, speakers, and duration before export. 2. Install and validate MFA and language-specific pronunciation resources. 3. Run preprocessing and forced alignment for each language, manually inspect alignment samples, and extract mel, pitch, energy, and duration targets. 4. Train both models on a CUDA-capable machine, preserving logs and checkpoints. 5. Synthesize the exact assignment test sentences, generate the required plots, and complete the final report from measured results. 6. Create and verify a public GitHub repository containing code and documentation only; do not publish restricted audio or credentials.

Reproduce the completed split audit

From the project root in PowerShell:

```
python tools\audit_manifests.py --manifest data\Svarah\manifest.jsonl --audio-root data\Svarah --output-dir data\Svarah\splits --language English
```

The workspace is local at `C:\Users\manso\OneDrive\Documents\ChatGPT\Fast2Speech`. Git currently has no remote configured, so there is no public repository link to provide yet.

References

1. FastSpeech 2 assignment handout supplied by the student, *FastSpeech2_Assignment.pdf*.

2. ming024/FastSpeech2 reference implementation, commit d4e79eb52e8b01d24703b2dfc0385544092958f3. The upstream README reports under 10,000 steps (under one hour on a GTX 1080 Ti) to generate acceptable-quality samples; this is not a runtime guarantee for this project or its configurations. github.com/ming024/FastSpeech2.

3. AI4Bharat, Svarah dataset card and terms. huggingface.co/datasets/ai4bharat/Svarah.

4. SPRINGLab IndicTTS Telugu dataset card. Its description includes conflicting language information; use only after verification. huggingface.co/datasets/SPRINGLab/IndicTTS_Telugu.

Limitations: This report records project status and a dataset split audit only. The audit does not prove transcript accuracy or alignment. No model metrics or generated outputs exist yet.